The Global Gender Gap in STEM Applications: Pipeline vs. Choice

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Abstract

Women make up only 35% of STEM graduates worldwide. We use ranked applications and assignments from ten centralized university admissions systems to distinguish gender differences in academic preparation (the pipeline) from differences in application decisions (the choice gap). Among students in the top 10% of each setting's academic performance distribution, women are about 24 percentage points less likely than men to rank STEM first, averaging across settings. The pipeline gap varies widely and broadly tracks wage parity; the STEM choice gap remains large and negative everywhere. Linking applications to assignments shows that women's underrepresentation largely precedes allocation. Pooled across settings, women are 51.2% of high achievers, 38.6% of those listing STEM anywhere, 34.0% of those ranking it first, and 34.7% of those assigned to STEM. Holding application rates fixed, a gender-balanced pipeline would still leave high-achieving STEM applicants between 57% and 75% male in every setting. A strong female academic pipeline can therefore coexist with substantial underrepresentation in STEM.

Keywords: gender inequality, STEM gender gap, centralized application platforms
JEL: I23, I24, J16, D47

1. Introduction

Despite decades of progress in educational attainment, women remain substantially underrepresented in science, technology, engineering, and mathematics (STEM). In 2024, they accounted for only 35% of STEM graduates worldwide, a share that has barely moved in a decade (UNESCO, 2024). Explanations for this persistence are typically grouped into two channels: a *pipeline* channel, gender differences in academic preparation and access to selective STEM programs (Card and Payne,

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2021, Aucejo and James, 2021, Humphries et al., 2023), and a *choice* channel, gender differences in field preferences conditional on eligibility (Zafar, 2013).

Field of study matters for students' later outcomes (Altonji et al., 2016). Exploiting admission cutoffs, Kirkebøen et al. (2016) and Hastings et al. (2013) show that the field of specialization substantially affects later earnings, and that selective STEM programs are among those with highest returns. STEM fields are also central to innovation, productivity and growth (Toivanen and Väänänen, 2016, Peri et al., 2015). Thus, persistent gender gaps in STEM entry matter for both equity and efficiency as they keep women out of some of the highest-return careers and leave part of the available talent unused where it may be most productive (Weinberger, 1999, Hoogendoorn et al., 2013, National Science Foundation, 2017, Hsieh et al., 2019).

Distinguishing pipeline from choice is empirically difficult, since it requires observing both program-specific eligibility and students' ranked application decisions. While Delaney and Devereux (2019, 2021) demonstrate the value of using application data to study gender differences in college applications and STEM choices, existing research has largely been confined to single-country settings, limiting our ability to assess whether these patterns generalize across distinct institutional and cultural contexts.

We use administrative microdata from centralized admissions systems in ten settings across five continents: Australia, Brazil, Chile, China, Finland, Greece, Spain, Sweden, Taiwan, and Uganda. Despite large differences in population size, economic development, and gender norms, these systems share a key institutional structure: students submit ranked preferences over college–major combinations through coordinated platforms running deferred acceptance (DA) (Neilson, 2024), and are assigned to the highest-ranked option for which they are eligible. Eligibility depends almost exclusively on academic performance. This setting allows us to observe students' ranked choices while comparing women and men with similar admission prospects.

Within this setting, we focus on high-achieving students, defined as those in the top 10% of each setting's academic performance distribution, and measure STEM preference by whether a student ranks a STEM program first on the application list. Under DA, the first choice is the program a student prefers to every other option and the one they are assigned to whenever they are eligible, so ranking STEM first captures both the intensity of STEM preference and the margin that matters most for admission outcomes. Truthful ranking is optimal when lists are unconstrained (Roth, 1982), and although list-length limits vary widely across our settings, they do not predict the size of the gaps we find (Section 2).

Online Appendix B complements this analysis in two directions. It changes the definition of high

achievement, repeating the analysis for the top 5%, 20% and 30% and across the full performance distribution, and it uses a broader measure of STEM preference, whether a student lists a STEM program anywhere on the list, which captures interest that other options outrank. The choice gap remains large and negative under both changes.

Women's underrepresentation in STEM largely precedes allocation (Figure 4). Pooled across settings, women are 51.2% of high achievers, 38.6% of those listing STEM anywhere, 34.0% of those ranking it first, and 34.7% of those assigned to STEM. Assignment does not further reduce women's pooled share relative to first-choice STEM applicants.

Women are substantially underrepresented among high-achieving STEM applicants, and we decompose that underrepresentation into the pipeline and choice gaps defined in Section 3. Pipeline patterns vary markedly across settings and in sign, at every threshold we examine, yet women remain underrepresented even where they are a larger share of top students than men. The pipeline alone cannot account for what we observe.

By contrast, the choice gap is consistently large and negative. Among students in the top 10%, women are 23.7 percentage points less likely than men to rank a STEM program first, averaging across the ten settings, or 22.4 points pooling students; the gap exceeds 20 points in eight of the ten systems and holds under both measures everywhere. The choice gap also changes little with the definition of high achievement: pooled across settings, the gap moves by less than one percentage point as that group widens six-fold, from the top 5% to the top 30%.

Figure 3 shows how pipeline and choice combine: settings with greater female representation among high achievers tend to have larger male-to-female ratios of STEM application rates, which partially offsets the cross-setting differences in pipeline composition. Figure 5 places this pattern in its labor-market context: greater wage parity is associated with a stronger female academic pipeline, yet the STEM choice gap remains large and negative across the full range of wage parity. High wage parity and a female-majority pipeline can coexist with substantial female underrepresentation in STEM applications.

To our knowledge, this paper provides the first cross-national decomposition of the STEM gender gap into a pipeline gap (access and preparedness) and a choice gap (application decisions), using harmonized application microdata. We build on two strands of research. A first strand disentangles pipeline and choice within single settings (Delaney and Devereux, 2019, Card and Payne, 2021), but a single setting cannot show whether the pattern generalizes. A second strand, typically in the form of international reports (OECD, 2017, Encinas-Martín and Cherian, 2023, UNESCO, 2024), documents STEM gender gaps across education systems but cannot separate pipeline from choice, since it relies

on enrollment outcomes that blend pipeline, choices, and admissions. We instead assemble a global dataset of ranked application lists, which records the programs students put themselves forward for rather than only where they ultimately enroll.

DA platforms allow us to compare applications under similar admissions rules. In decentralized systems, applications combine preferences with beliefs about admission chances, application costs, and portfolio strategy. Coordinated platforms reduce these complications: students submit rankings and eligibility depends primarily on academic performance. Comparing high-achieving women and men separates observed application choices from the pipeline and limits several alternative explanations, without identifying a preference parameter.

The data do not identify the mechanisms behind the gap. Residual strategic incentives remain where lists are short or where applications precede scores, which we take up in Section 2. Even students who report truthfully may have different information: women and men may hold different beliefs about what studying and working in STEM involves, may face or expect different returns to a STEM degree (Zafar, 2013, Wiswall and Zafar, 2018), or may search and learn less about STEM options before applying, so that the list they submit faithfully reflects a choice set they never fully explored (Agte et al., 2025). The admissions rules help us assess strategic explanations, but the records do not reveal the beliefs behind students' rankings.

2. Data

This section outlines the institutional context and data for the ten settings we study: Australia, Brazil, Chile, China, Finland, Greece, Spain, Sweden, Taiwan, and Uganda.[1] Our data are nationwide in eight of the ten settings. In Australia we observe Victoria, a state holding 25% of the population and a strong proxy for the nation; in China, Ningxia, a more rural and less developed autonomous region holding 0.5% of the population.

Institutional coverage is broad across our settings. For our analysis, the platforms' role in allocating seats at selective programs matters more than the number of institutions they include. In eight of the ten settings the centralized platform is the route into the most selective universities and their STEM programs, and where participation is partial the institutions outside it are typically less selective. Two settings depart from this. In Brazil the São Paulo state universities, which host some of the country's strongest STEM programs, run their own admissions, so the Brazilian sample

[1] Online Appendix A provides additional details on each setting.

covers the federal system rather than the full public sector. In Uganda the platform allocates only government-sponsored places, roughly a tenth of undergraduate seats at Makerere, the largest public university, with the remainder filled through privately sponsored admissions that each university runs itself, so the Ugandan sample covers the government-sponsored track rather than the selective sector as a whole. Elsewhere the same logic governs which students we observe: where selective seats are allocated centrally, a student who wants one must appear on the platform, so covering the top of the institutional distribution implies covering the top of the student distribution.[2]

The ten settings differ substantially in population, income, and inequality: population ranges from 5.5 million in Finland to 1.4 billion in China, GDP per capita from about $4,000 in Uganda to $66,000 in Taiwan, and the Gini index from 27.3 in Finland to 53.9 in Brazil. To characterize the labor-market context we use a wage parity index, defined as one minus the total gender wage gap, which ranges from 0.517 in Uganda to 0.965 in Greece; its sources are described in Online Appendix G and its relationship with the gaps in Section 4.6.[3]

Despite these differences, the systems share two features: they allocate most university seats through centralized platforms running variants of deferred acceptance, and eligibility is almost exclusively based on academic performance, typically high school grades and admission exam scores. Our data, drawn from the admissions agencies, include students' ranked applications, their performance records, and the program each is assigned to.[4]

Centralized admissions cover every university in Australia, Greece, and Sweden, and a subset elsewhere, from 36 of 38 in Finland down to 8 of 52 in Uganda and 132 of 2,448 in Brazil, where the 2,152 private institutions and the state university systems admit separately. Four settings charge no tuition among participating universities; financial aid is available in all ten.

Students apply to specific college–major combinations, choosing from 149 options a year in Uganda and more than 18,000 in China, and ranking at most two of them in Brazil against no numerical limit in Greece. All are DA mechanisms, which give students little incentive to misreport and so let us read the submitted list as an ordering over the programs a student was willing to enter. This common structure limits strategic explanations for differences in submitted rankings, subject to the restrictions discussed below.[5]

[2]The systems differ in how much of the wider cohort they capture, with Spain's entrance exam taken by 81% of a cohort and Uganda's pool smaller and more selected, but the choice gap is negative throughout the performance distribution in every setting, not only at the top; Online Appendix B reports the gap by score bin.

[3]Online Appendix Table A.I reports these and the admission-system characteristics summarized below.

[4]Because the periods covered differ across settings, Online Appendix D re-estimates on the cohort closest to 2018 in each; the results are very similar.

[5]Strategy-proofness weakens when the list is restricted (Fack et al., 2019), and list censoring is a natural concern

They also vary in scale, from 36.2 thousand yearly applicants in Uganda to 2.71 million in Brazil. Women outnumber men among applicants in eight of the ten settings, from 53.0% in Chile to 60.1% in Finland; the exceptions are Taiwan at 49.8% and Uganda at 43.9%. These patterns mirror findings for the United States (Goldin et al., 2006). The cohorts we observe range from 2003–2012 in Greece to 2020–2024 in Spain; Uganda contributes one academic-year cohort, 2017/2018 (entry in fall 2017).

Eligibility also depends on prior subject choices in Greece and Uganda. Greek lists are unlimited in length but confined to two of five broad subject categories, so an applicant chooses a domain before ranking within it. In Uganda, a student whose A-level subject combination is non-STEM cannot apply to a STEM program at all, and that combination is fixed years before the application. In both settings, part of what we measure as choice is a track decision taken earlier, and in Uganda it is a decision that forecloses the option rather than merely discouraging it. These restrictions limit the interpretation of the choice gap. However, Uganda has the smallest choice gap in our data and Greece sits mid-range, so neither setting is generating the pattern we report.[6]

The settings differ in other ways too—exam design, score scales, tested content—and we do not attempt to decompose how they contribute to cross-country differences in either gap. Within each setting, we compare women and men ranked on the same performance measure under the same rules to assess whether a gap in STEM applications persists among students facing similar admission prospects.

3. Empirical strategy

Eligibility in these centralized systems depends on academic performance, so students with similar performance face similar admission prospects. We therefore focus on high-achieving students, defined as those in the top 10% of their cohort on the mandatory components of each setting's performance measure (university admission exam, high school exit exam, or high school GPA), which lets us include all applicants regardless of intended field. These are the students most likely to gain admission to and succeed in selective STEM programs. The cutoff is a reporting convention. Online

where the cap is tight, as in Brazil. Across our settings, however, the caps do not order the gaps: counted in college–major cells they run from two in Brazil and 24 in China to a hundred in Taiwan, with no numerical limit in Greece (Online Appendix Table A.I), yet the smallest choice component is Uganda's, on a six-cell list, the second smallest Taiwan's, on a hundred-cell list, and Brazil's two-cell list sits mid-range. We read this as evidence against list length as the main driver, without excluding it in any single setting. Brazil also operates an iterative DA in which students revise across rounds, and the evidence on how well iterative designs sustain truthful reporting is mixed (Bó and Hakimov, 2020, 2022, Barahona et al., 2023).

[6] Online Appendix A gives both in full. A parallel restriction operates through prerequisites in several other settings, but only Greece and Uganda impose a categorical gate.

Appendix B repeats every exercise at the top 5%, 20%, and 30% and across the full distribution, and the choice gap is essentially unchanged. Thresholds are computed within cohort and system, so students are never ranked against a cohort other than their own.

We define STEM using ISCED-2013 two-digit fields of education and training,[7] grouping natural sciences, mathematics and statistics (05), information and communication technologies (06), and engineering, manufacturing and construction (07). Each system's programs were mapped to these fields using the official national crosswalk, and the harmonized counts were tabulated for STEM as a whole.

Our analysis begins by characterizing the gender composition of high-achieving STEM applicants. We then decompose these gender differences by examining two gaps:

1. The pipeline gap: difference between women's and men's representation among students in the top 10% of each setting's academic performance distribution.

2. The choice gap: difference in the share of high-achieving women and men who rank a STEM program either as their top choice or at any rank of their preference list.[8]

These are not competing explanations of female underrepresentation but components of a single accounting identity. The number of high-achieving applicants to a field is the product of two margins: how many students of each gender reach the top group, and what share of them rank the field first. Because it is a product, taking logs makes the two margins additive, so the gender gap in applicants splits exactly into the two gaps defined above, with no residual.

Formally, let T_g denote the number of students of gender $g \in \{m, f\}$ in the top group and s_g the share of them who rank a given field first, so that $A_g = T_g\, s_g$. Taking logs of the ratio between genders gives

$$\underbrace{\ln\left(\frac{A_m}{A_f}\right)}_{\text{gap among high-achieving applicants}} = \underbrace{\ln\left(\frac{T_m}{T_f}\right)}_{\text{pipeline component } P} + \underbrace{\ln\left(\frac{s_m}{s_f}\right)}_{\text{choice component } C}. \tag{1}$$

Note the orientation: the two gaps defined above are stated female minus male, so that a negative choice gap means women apply less often, whereas P and C are stated male relative to female, so

[7] See UNESCO Institute for Statistics, *International Standard Classification of Education (ISCED)*: https://uis.unesco.org/en/topic/international-standard-classification-education-isced. Business and law (code 04) and health (code 09) serve as comparison fields throughout; Online Appendix C reports those results in full.

[8] List-composition statistics, such as the STEM share of a list or a rank-weighted index, are not comparable across systems whose list-length rules range from two programs (Brazil) to no limit (Greece).

that positive values favor men throughout the decomposition. The male share among high-achieving applicants is $\Lambda(P+C)$ with $\Lambda(z) = 1/(1+e^{-z})$, which yields the counterfactual we use below and in Online Appendix B: $\Lambda(C)$ is the male share that would obtain if the top group were gender balanced, and $\Lambda(P)$ the share if men and women applied at equal rates, each holding the other margin at its observed value.

The components are directly comparable, whereas the underlying percentage-point gaps measure different quantities: composition shares for the pipeline gap and application rates for the choice gap. Since P depends only on the composition of the top group, it is common to every field within a setting and any cross-field difference is by construction attributable to C.[9] We also relate both gaps to cross-country wage parity, a labor-market measure directly relevant to specialization decisions.

The choice component compares men and women within the top group rather than at a common score, so where men sit higher within it and application rates rise with performance, part of C reflects within-group composition. We assess sensitivity to the width of this group by narrowing the performance threshold. Recomputed on the top 5%, C remains positive in all ten settings, from +0.28 to +1.08, and on the top 1% in all nine settings for which that group can be constructed, from +0.13 to +0.80. Across those nine settings the mean falls from +0.69 at the top 10% to +0.55 at the top 1%, and it is lower at the top 1% in eight of the nine. The application gap therefore persists even within the narrowest high-achieving groups we observe. These threshold comparisons do not isolate the contribution of within-group score composition.

4. Results

4.1. Female representation among top STEM applicants

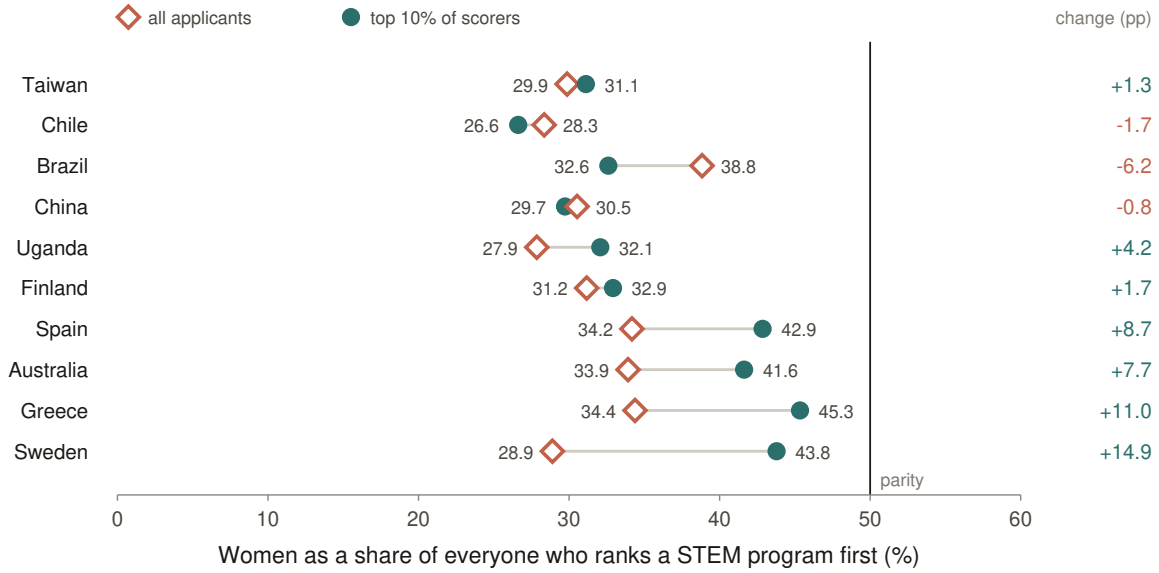
Figure 1 reports, for each system, women's share among STEM applicants, defined as students who rank a STEM program first. It shows this share among all applicants and among students in the top 10% of the academic performance distribution. As discussed in the introduction, ranking STEM first is the most informative measure of STEM preference under DA and bears directly on admission outcomes. Online Appendix B replicates the main results under the broader any-choice definition, which counts a STEM program listed anywhere on the application list.

Women are underrepresented among first-choice STEM applicants in every system. Among all applicants, their share ranges from 27.9% in Uganda to 38.8% in Brazil. One might expect this gap

[9] Online Appendix E extends the identity to realized assignments and reports the additional component.

to narrow among the strongest students, who are the most likely to be admitted to selective STEM programs, but it does so only partly. Restricting attention to the top 10% raises women's share in seven of the ten systems, yet in six of them it remains below 35%, and in the other four it reaches at most 45.3%. This pattern persists under alternative definitions of high achievement.[10] Under the broader any-choice definition, which counts STEM programs listed anywhere on the application, women's share among top-10% STEM applicants rises in every setting, but women remain a minority in nine of the ten.[11]

Figure 1: Women as a Share of STEM Applicants, All Applicants and Top 10%



Notes: Women as a percentage of everyone ranking a STEM program first, among all applicants (diamonds) and among the top 10% of the academic performance distribution (circles). The right-hand column gives the change, in percentage points. Samples in Online Appendix Table A.II.

The shares in Figure 1 combine the two forces of equation (1): the *pipeline gap*, women's representation among high-achieving students overall, and the *choice gap*, the propensity of high-achieving women to rank STEM relative to high-achieving men. We analyze each in turn.

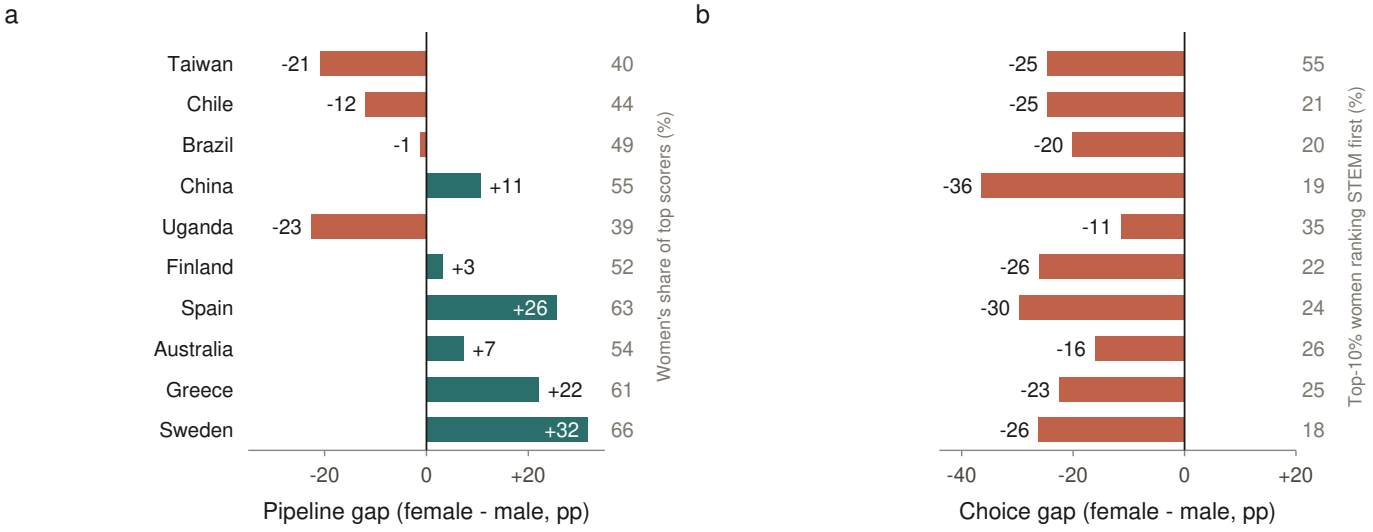
[10] Online Appendix B reports the results for the top 5% and the top 20%.
[11] Online Appendix B.6 reports the any-choice results at all three thresholds.

4.2. The pipeline gap

Panel (a) of Figure 2 reports the *pipeline gap*, the female minus male share of students in the top 10% of the academic performance distribution; the column beside each setting's name gives women's share of that group.[12] It varies widely across our settings, and in sign: women are underrepresented among high-achieving students in Taiwan, Chile, and Uganda, and overrepresented in Australia, China, Greece, Spain, and Sweden; Brazil is balanced at every threshold and Finland changes sign with the threshold. Uganda and Sweden mark the extremes, at 20 to 25 percentage points below parity and 31 to 33 above.

The pipeline gap therefore cannot account for the pattern in Figure 1. In every setting, and under either definition, women's share among high-achieving STEM applicants falls below their share among high-achieving students overall, including in the settings where they are the majority of that group.

Figure 2: The Pipeline Gap and the Choice Gap among Top 10% Students



Notes: Female minus male, percentage points, top 10% of the academic performance distribution: (a) the pipeline gap, (b) the choice gap. Gray columns give the female level. The two are differences in different quantities and do not sum; Online Appendix E gives the components that do. For Chile, panel (b) conditions on the top-decile students who submit an application list, while panel (a), Online Appendix Tables B.I and C.I, and the decomposition components use the full top-decile totals; that convention moves Chile's choice gap from −23.1 to the −24.6 plotted here and leaves every other setting unchanged. Samples and denominators in Online Appendix Table A.II.

[12] Online Appendix B.4 reports the same gap at the top 5% and top 20%, which we draw on below.

4.3. The choice gap

Following Delaney and Devereux (2019), we define the *choice gap* as the difference between high-achieving women and men in the probability of ranking a STEM program first. Panel (b) of Figure 2 reports this gap for students in the top 10% of the academic performance distribution, on the same scale as the pipeline gap in panel (a). The column beside each setting's name gives the share of top-10% women who rank STEM first, from which men's rate follows by subtracting the gap. In contrast to the wide variation in the pipeline gap, the choice gap is negative in every setting. High-achieving women are less likely than men to rank STEM first by between 11.5 percentage points in Uganda and 36.5 points in China, and the gap exceeds 20 points in eight of the ten settings. This is consistent with single-country evidence from Chile, which documents a similar gap in major choice within a centralized system (Bordón et al., 2020).

The choice gap remains negative under alternative definitions of high achievement and of STEM preference. It is negative in every setting at the top 5%, 10%, 20% and 30%, and the pooled gap varies by less than one percentage point between the narrowest and the broadest of these groups.[13] When we count STEM programs listed anywhere on the application, the gap is also negative in every setting and at every threshold, and it exceeds 20 percentage points in seven of the ten settings at the top 10%.[14]

Two features of the admission systems bear on how we interpret this gap. First, admission rules are not sex-neutral everywhere. The performance measure on which we rank students is constructed identically for women and men in every setting, but Uganda gives female applicants a 1.5-point bonus on the official score, reserves 40% of STEM places at the largest public university for women, and sets separate STEM cutoffs for men and women (Online Appendix A.1). Such rules can affect submitted rankings only if students anticipate them, and anticipating a female advantage should make women more willing to rank STEM, narrowing the gap. Uganda indeed has the smallest choice gap in our sample.

Second, students may rank strategically when lists are constrained (Fack et al., 2019). This is unlikely to drive our results. List length does not predict the size of the gap across settings (Section 2), and the gap persists throughout the performance distribution, including among lower-scoring

[13] Online Appendix Table B.I reports these figures setting by setting. The margin that does move systematically with the threshold is admission rather than application. The gap among students admitted to STEM narrows from 18.2 to 14.2 percentage points between the top 5% and the top 30%, as cutoffs bind on a larger share of a wider group.

[14] Online Appendix B reports both definitions at all thresholds and across the full performance distribution. Under the any-choice definition the gap is smaller on average across settings, though marginally larger when students are pooled.

students, for whom the risk of rejection makes strategic omissions most likely.[15] We therefore read the gap as reflecting differences in application choices, while recognizing that submitted rankings may retain some strategic component.

Cross-field comparisons sharpen the distinction between pipeline and choice. The pipeline component is common to every field within a setting (Section 3), while the choice component differs markedly across fields (Online Appendix Table C.I). It is positive for STEM in all ten settings but negative for health in eight: high-achieving women are more likely than men to rank health first. The two health exceptions, Greece and Taiwan, have gaps of less than two percentage points; under the any-choice measure, the female advantage in health holds in all ten settings. Business and law shows no systematic sign, with a positive component in four of nine settings.

Holding field-specific application rates fixed, a gender-balanced pipeline would leave STEM applicants between 57% and 75% male, compared with 34% to 52% for health. The ranges do not overlap. These comparisons hold the applicant pool, performance threshold, admission mechanism, and list-length constraint fixed within each setting. The persistent female shortfall is specific to STEM, rather than a general feature of selective, high-return fields.[16]

4.4. How pipeline and choice combine

Figure 3 plots the pipeline and choice components of equation (1) for each setting. Settings with a stronger female pipeline tend to have a larger male advantage in STEM application rates, and the two components are negatively correlated (-0.65).[17]

Both components are male-to-female log ratios, so positive values favor men. Because they add up to the log odds that a high-achieving STEM applicant is male, each gray diagonal joins the combinations of P and C that produce the same male share among first-choice STEM applicants, $\Lambda(P + C)$, from 55% to 75%. Settings on the same diagonal have equally male applicant pools, whatever the mix of pipeline and choice behind them, and moving right or up shifts a setting toward a more male pool.

The diagonals show three things. First, every setting lies at or above the 55% diagonal, including those where women are the majority of high achievers. In those settings, the choice component more than offsets women's advantage in the pipeline. Second, because the two components move in

[15] Online Appendix B reports the gap by score bin.

[16] Online Appendix C reports the comparison across fields and thresholds; the project's online materials provide the underlying probabilities and counts.

[17] The correlation remains negative when any single setting is omitted and at every threshold we examine.

opposite directions, settings that differ widely in their pipelines end up closer together across the diagonals. Women's share spans 27 percentage points across settings among top-decile students but only 19 points among first-choice STEM applicants. Third, moving a setting horizontally to $P = 0$ gives the male share it would have with a gender-balanced pipeline, $\Lambda(C)$, which ranges from 57% to 75%. The pipeline still matters, since for a given C, a lower P reduces the male share of applicants.

The cross-setting correlation does not identify how choices would respond to a change in preparation, nor why the two components covary. Two explanations are consistent with the data. The first is composition. Where more women reach the top group, the additional women may have a comparative advantage outside mathematics-intensive fields, which would lower the female STEM application rate. Section 3 shows that C shrinks but does not vanish as the group narrows to the top 1%. The second is context. The three settings with the strongest female pipelines, Greece, Spain, and Sweden, are also among those with the highest wage parity (Section 4.6), the configuration in which the gender-equality-paradox literature reports larger gender differences in field choice.

4.5. From applications to assignments

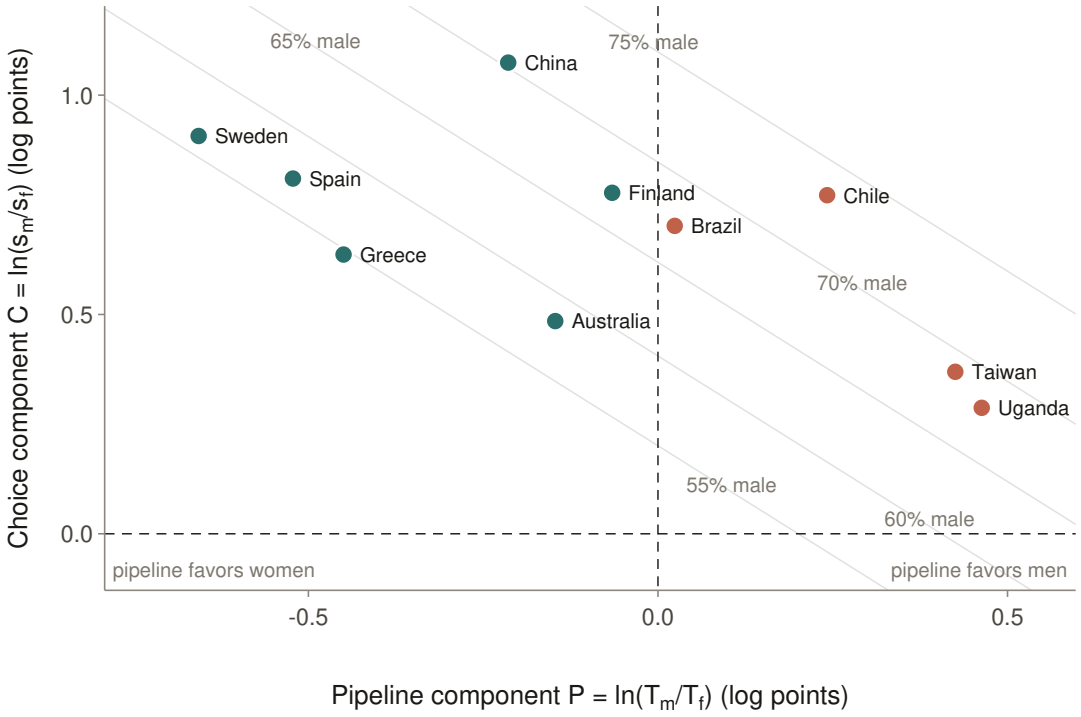
The application shortfall largely carries through to assignment. Figure 4 compares four groups of high achievers. Pooling students across the ten settings, so that larger systems carry more weight, women are 51.2% of the top decile, 38.6% of those listing STEM anywhere, 34.0% of those ranking it first, and 34.7% of those assigned to STEM.[18] The largest compositional difference is between the top decile and those listing STEM at all. The closeness of the final two shares indicates that most of the aggregate shortfall precedes allocation, rather than emerging as an additional net loss at assignment.

These are group compositions, not individual transition or acceptance rates: students can be assigned to a STEM program ranked below first. The assignment component V is smaller in magnitude than C in nine settings and comparable in Uganda, with mixed signs and an unweighted mean of +0.005 log points.[19] A small net component can mask offsetting assignment flows; it does not establish equal treatment of otherwise identical applicants or rule out anticipated admission constraints shaping their lists. Assigned programs may also differ from actual enrollment.

[18] Brazil and Chile supply two thirds of the pooled top decile. The unweighted means across the ten settings are 52.2%, 41.8%, 35.9%, and 35.8%, the same pattern.

[19] Online Appendix E defines V and Online Appendix F reports its magnitude; Online Appendix B.8 reports the accounting counterfactuals.

Figure 3: Pipeline and Choice Components of the STEM Gender Gap



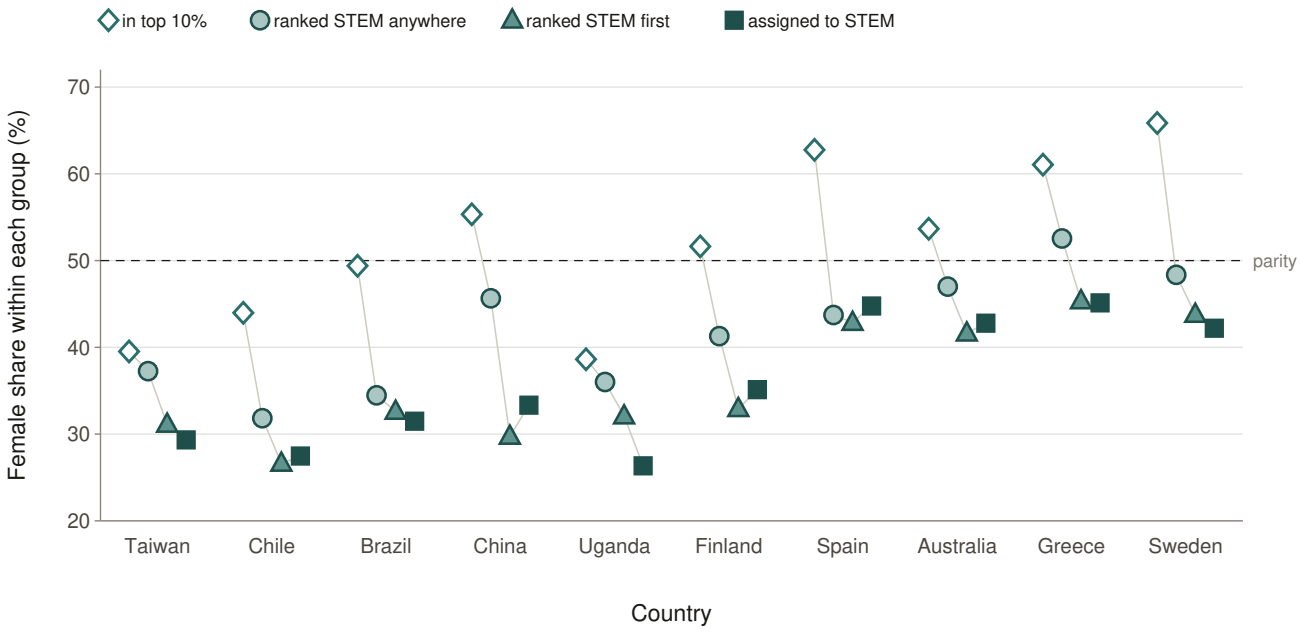
Notes: One point per setting, top decile. $P = \ln(T_m/T_f)$ measures pipeline composition and $C = \ln(s_m/s_f)$ the ratio of first-choice STEM application rates. Unlike the female-minus-male percentage-point gaps in Figure 2, these are male-to-female log ratios: positive values favor men. Gray diagonals have slope -1 and hold $P + C$, hence the male applicant share $\Lambda(P + C)$, constant. Moving right or up increases that share. These are accounting contours, not fitted lines. Marker color follows the sign of P: teal where women outnumber men among high achievers, red where men do. Dashed lines mark $P = 0$ and $C = 0$. Samples in Online Appendix Table A.II.

4.6. Wage parity and the persistence of the choice gap

Figure 5 relates both gaps to wage parity, defined as one minus the total gender wage gap, with both panels on the same female-minus-male percentage-point scale. The two gaps relate to wage parity very differently. Women's representation among high achievers rises steeply with wage parity (panel a). Across settings, a 0.1 increase in wage parity is associated with an 11 percentage-point increase in the pipeline gap in women's favor. The choice gap, by contrast, is nearly flat (panel b). The same increase is associated with a change of about two points, which is not statistically different from zero and reverses sign when Uganda is excluded.[20] These are correlations across ten settings, and they do not identify the effect of wage parity on either gap.

[20] Online Appendix G reports the wage sources and sensitivity checks. Expressed as log ratios, the choice component rises with wage parity ($t = 2.0$), but the association is weak once Uganda is excluded ($t = 0.7$).

Figure 4: Female Representation among High Achievers, STEM Applicants, and STEM Assignees, Top Decile

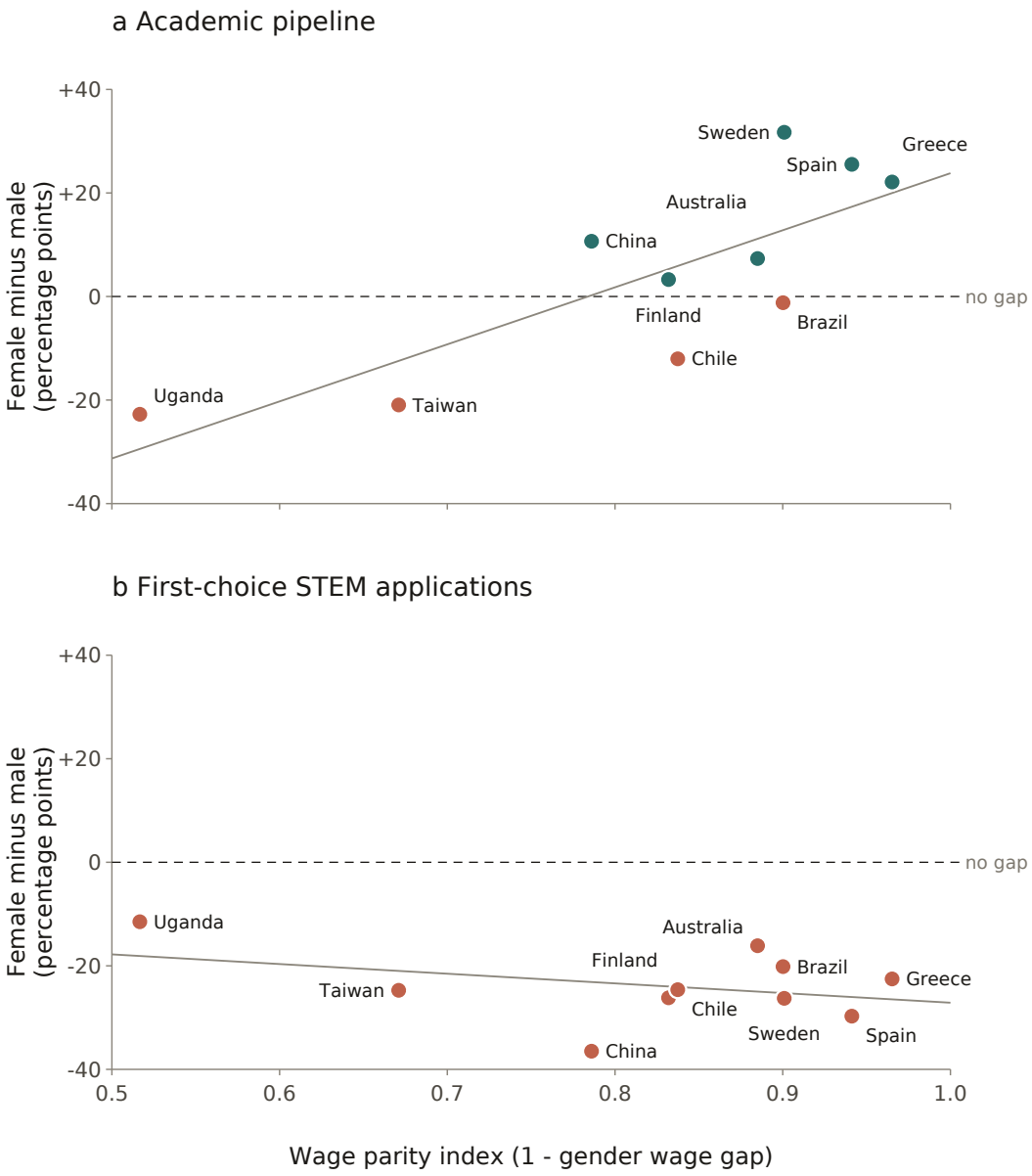


Notes: Female share within each indicated group of top-decile students, not an individual transition or acceptance rate. The assignment group need not be a subset of first-choice STEM applicants: students can be assigned to STEM from lower ranks. Differences between markers describe aggregate composition, not causal effects of admissions. Online Appendix B.8 reports the counterfactuals; sample definitions are in Online Appendix Table A.II.

The flat slope in panel (b) means that the female disadvantage in STEM applications is present across the full range of wage parity. In Greece, Spain, and Sweden, women are a clear majority of high achievers and wage parity is high, yet high-achieving women remain 23 to 30 percentage points less likely than men to rank STEM first. A strong female pipeline and high wage parity thus coexist with a substantial STEM application gap.[21]

[21] This pattern relates to the gender-equality paradox discussed by Stoet and Geary (2018, 2020), Richardson et al. (2020), Breda et al. (2020), and Falk and Hermle (2018).

Figure 5: Academic Pipeline and STEM Choice Gaps across Levels of Wage Parity



Notes: One point per setting, top decile. Both panels use the same female-minus-male percentage-point scale: (a) the difference in women's and men's shares of high achievers; (b) the difference in their first-choice STEM application rates. Panel (b) uses the denominators in Figure 2, including applicant conditioning for Chile. Higher wage parity means a smaller total gender wage gap. Solid lines are unweighted least-squares fits; the dashed line marks no gap. Marker color follows the sign of the plotted gap: teal where it favors women, red where it favors men. The choice-gap slope is sensitive to excluding Uganda. Wage sources and sensitivity checks are in Online Appendix G; sample definitions are in Online Appendix Table A.II.

5. Discussion and conclusion

We decompose women's underrepresentation in STEM into pipeline and choice gaps across ten centralized admissions systems. The pipeline varies substantially: women make up under 40% of high achievers in Uganda and over 60% in Sweden. Yet high-achieving women are less likely than men to apply to STEM in every setting, under both application measures and at every threshold we examine. Among top-10% students, women are about 24 percentage points less likely than men in the same high-achieving group to rank STEM first.

The application shortfall largely carries through to assignment. Pooled across settings, women are 51.2% of high achievers but only 34.0% of first-choice STEM applicants and 34.7% of those assigned to STEM. Most of the aggregate shortfall is already present before allocation. These shares describe group composition; a small net assignment component can conceal offsetting flows.

Holding application rates fixed, a gender-balanced pipeline would still leave high-achieving STEM applicants between 57% and 75% male in every setting. A stronger academic pipeline alone therefore need not translate into gender balance in STEM applications, consistent with evidence that pipeline factors cannot fully account for persistent underrepresentation (Patnaik et al., 2021).

The records do not identify the causes of the choice gap. Prior research points to differences in how students value job attributes and programs (Zafar, 2013, Wiswall and Zafar, 2018). Submitted rankings may also reflect beliefs about returns and discrimination, family considerations, identity and belonging, or incomplete search.[22] The stability of the gap across settings is not evidence of innate or biological differences in preferences, since application choices are formed within social, cultural and institutional environments. Its persistence may reflect drivers shared across settings, or different combinations of context-specific forces that produce gaps of similar size, and our data cannot distinguish between the two. Linking data on beliefs and search to application records could help explain how the persistent STEM application gap forms, including where women already constitute a majority of high achievers.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The analysis uses individual-level administrative records from the ten centralized admission systems listed in Online Appendix Table A.II. These records are confidential and were obtained under separate agreements with each agency, so we are not able to redistribute them; researchers seeking access should apply to the agency named for that setting in the same table. The aggregated data underlying every figure and table carry no such restriction: for each setting, performance threshold and outcome we report the numerator, the denominator and the resulting rate, and these, together with the code that draws the exhibits from them, are available in the project's online materials, at https://gendergapinsights.com.

The Global Gender Gap in STEM Applications: Pipeline vs. Choice

Online Appendix

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Appendix A. Institutional details

This section documents the ten admission systems and the data we draw from each. Table A.I collects the setting, university-system and admission-system characteristics summarized in Section 2. Table A.II gives the administrative source, cohorts, sample size, and gender composition of the top decile—the group the main-text analysis is based on. Table A.III then sets the ten systems side by side on the features that bear most directly on how we read a submitted list: the score students are ranked on, whether they know that score when they apply, how places are assigned, and whether any admission rule differs by sex.

Fuller per-setting descriptions (each system's application calendar, the institutions inside and outside the centralized platform, and how the analysis sample was constructed) are available in the project's online materials at https://gendergapinsights.com.

Table A.I: Institutional characteristics

	Australia (1)	Brazil (2)	Chile (3)	China (4)	Finland (5)	Greece (6)	Spain (7)	Sweden (8)	Taiwan (9)	Uganda (10)
Panel A										
Setting Characteristics										
Population (millions)	24.6	209.5	18.7	1,402.8	5.5	10.7	46.8	10.2	24.0	40.1
GDP per capita (thousands)	56	21	31	24	56	41	50	62	66	4
Human Development Index	0.937	0.764	0.849	0.755	0.937	0.881	0.905	0.943	0.925	0.534
Gini index	33.7	53.9	44.4	38.5	27.3	32.9	34.7	30.0	34.2	42.8
Wage parity index (1− wage gap)	0.885	0.900	0.837	0.786	0.832	0.965	0.941	0.901	0.671	0.517
Panel B										
University System										
N of Institutions	21/21	132/2,448	34/60	1,252/2,663	36/38	41/41	50/86	41/41	65/71	8/52
Tuition fees	Yes	No	Yes	Yes	No	No	Yes	No	Yes	Yes
Financial aid	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel C										
Admission System										
Options available (yearly avg.)	1,078	6,310	1,423	18,671	1,458	609	2,169	15,374	1,659	149
Max. apps	12	2	10	24	6	No limit	12	20	100	6
N of applicants in a year (avg.)	38.8	2,711.0	249.7	52.4	69.6	68.0	171.6	108.2	72.1	36.2
N of admitted students in STEM (avg./year)	7.9	65.0	30.7	18.9	8.1	17.9	32.6	13.6	27.7	1.3
Female share (apps)	55.8%	57.1%	53.0%	55.5%	60.1%	55.8%	56.8%	57.4%	49.8%	43.9%
Data coverage	2009/10	2016	2012/18	2018	2016/20	2003/12	2020/24	2017	2010	2017/18

Notes: Panel A describes each setting, panel B its university system, and panel C its admission system. Australia is Victoria and China is Ningxia. "N of Institutions" gives those in the centralized system over the total. "Max. apps" counts college–major combinations; in China this is the first undergraduate batch, where a student ranks four institutions with six majors in each, and the cap applies within a batch rather than across the cycle. Applicant and admitted-student counts are yearly averages, in thousands; the applicant row counts the score holders who form each setting's performance distribution, which in Chile is DEMRE's registered candidates, about half of whom go on to submit an application. Performance groups are formed within setting and cohort; Table A.III gives each setting's score measure, and the setting-specific notes that follow record where the groups cannot be filled exactly. Wage sources are described in Online Appendix G; Section 4.6 relates wage parity to the two gaps.

Appendix A.1. Setting-specific notes

Uganda. Admission rules here are the most explicitly sex-conditional in our sample, and they favor women. Female applicants have received an additional 1.5 points on the official weighted score since 1990. For the 2017/2018 cohort studied here, a 30:70 minimum STEM ratio applied at Busitema, alongside separate published cutoffs. Makerere's 40:60 female-to-male STEM policy began in 2020/2021, after our sample period, and should not be read as an admission rule for this cohort. None of this enters the performance measure we use. We rank Ugandan applicants on an index we construct ourselves from raw national-examination grades, which is identical in construction for women and men and carries no adjustment; the official PUJAB weighted score, which does carry the adjustment, is program-specific and is not used here. Because applications are submitted before results are released, these rules can reach the choice gap only through anticipation, and anticipating

Table A.II: Data sources and composition of the top 10% of the score distribution

				Students in the top 10%			
Country	Administrative source	Cohorts	N	Women	Men	Total	% Female
Australia	Victorian Tertiary Admissions Centre	2009–2010	77,648	4,158	3,590	7,748	53.7
Brazil	SISU platform / ENEM records	2016	2,711,049	133,899	137,150	271,049	49.4
Chile	DEMRE	2012–2018	1,747,925	76,534	97,486	174,020	44.0
China	Ningxia Department of Education	2018	52,427	2,879	2,324	5,203	55.3
Finland	Ministry of Education (HAREK, AMKOREK)	2016–2020	347,904	17,859	16,726	34,585	51.6
Greece	Ministry of Education	2003–2012	680,035	41,523	26,484	68,007	61.1
Spain	Ministry of Universities	2020–2024	857,870	53,783	31,908	85,691	62.8
Sweden	Swedish Council for Higher Education	2017	108,237	7,085	3,673	10,758	65.9
Taiwan	Joint Board of College Recruitment Comm.	2010	72,067	2,766	4,232	6,998	39.5
Uganda	Public University Joint Admissions Board	2017/18	36,247	1,231	1,956	3,187	38.6

Notes: Counts are the number of students in the top decile of each setting's academic performance distribution, from the harmonized administrative tables supplied by each country team. Deciles are defined over the full distribution of score holders in each setting, and N reports that full distribution (deciles 1–10, female + male). Australia corresponds to the state of Victoria and China to the Ningxia Autonomous Region. The top decile contains close to but not exactly 10% of score holders in Uganda (8.8%) and Taiwan (9.7%), because coarse score scales generate ties at the cutoff; Uganda's score is the integer sum of two national exam grades and takes only 17 distinct values. The final column reports women as a percentage of the top-decile total, the pipeline measure plotted in Figure 2. Unless a figure note states otherwise, these country samples, cohort years, and sample sizes apply to all figures in the main text and the Online Appendix; Online Appendix D instead uses the cohort closest to 2018 in each country.

an advantage would if anything push Uganda's gap toward zero—which is where Uganda sits, with the smallest choice component in our sample.

The performance measure is also unusually coarse. The index takes only seventeen distinct values, so cell sizes are uneven and ties mean the groups cannot be filled exactly: the top decile covers 8.8% of the cohort and the top 20% covers 14.7%, against 10% and 20% elsewhere. A true top-1% cutoff cannot be identified on a seventeen-value score, so where we report that group for Uganda it is the top score group, 547 students, or 1.5% of applicants. Interpolating between adjacent thresholds moves Uganda's gaps by under 0.03 log points.

Chile. Denominators differ here because the exam can be taken without submitting an application. Panel (b) of Figures 2 and 5, and the corresponding appendix figures, condition on top-decile students who submit an application list, while Table B.I, Table C.I and the decomposition components use the full top-decile totals reported in Table A.II. The convention moves Chile's top-10% choice gap from −23.1 to −24.6 percentage points and leaves every other setting unchanged.

Australia. More than one candidate ranking variant is available. We use the Australian Tertiary Admissions Rank throughout, the rank the clearinghouse itself publishes and allocates on; a variant ranking students on English and mathematics results only is not comparable with the measures used in the other settings and is not used here.

Greece. Field coverage is incomplete: counts for business and law are available only for the top 5%, so Greece is absent from those columns of Table C.I and is marked n/a in the top-10% and top-20% panels of the corresponding figures. Its STEM and health counts are complete.

Table A.III: Admission systems at a glance

Setting	Performance measure	Scores known when applying?	Assignment mechanism	Sex-conditional admission rules
Australia (Victoria)	ATAR, a percentile rank built from scaled subject scores	Yes	Clearinghouse; offer rounds from January	—
Brazil (federal)	ENEM score, weighted by program	Yes	Iterative DA; provisional cutoffs updated daily	—
Chile	High-school GPA and PSU sections, weighted by program	Yes	Deferred acceptance	—
China (Ningxia)	Gaokao total, within province and track	Yes	Parallel-choice DA within each batch	—
Finland	Matriculation grades and/or entrance exam	Partly; most applicants already hold a completed certificate	DA re-run over the results period	—
Greece	Panhellenic examination score	Yes	Score-ordered allocation along declared preferences	—
Spain	Bachillerato GPA and EBAU, weighted by program	Yes	Score-ordered regional assignment	—
Sweden	Upper-secondary GPA and/or scholastic aptitude test	No; the list closes before results are released	Rank-ordered assignment in two rounds	—
Taiwan	AST subject scores, weighted by program	Yes	Program-proposing DA	—
Uganda	Index of national-examination grades, constructed by us	No; applications precede results	Merit-order selection	+1.5 points for female applicants; 30:70 minimum STEM ratio at Busitema; separate published cutoffs

Notes: One row per setting, describing the system as it operated in the cohorts we observe (Table A.II); several of these systems have since been reformed. "Performance measure" is the score on which we rank students to form performance groups; in every setting it is constructed identically for women and men. "Scores known when applying" records whether a student's examination result is available at the moment the ranked list is submitted, which governs how much scope there is for admission-risk considerations to shape the list. "—" in the final column means we document no sex-conditional rule for that setting; we have not audited every program-level entry route in every system, and the entry for Uganda incorporates the country coauthor's clarification of the rules for the observed cohort. Institution counts, list-length caps, fees and cohort coverage are in Table A.I. Fuller per-setting descriptions, including application calendars and data construction, are available in the project's online materials.

Appendix B. Pipeline and STEM choice gaps along the academic performance distribution

The main text defines high-achieving students as those in the top 10% of each setting's academic performance distribution. That focus is motivated by the fact that these students are most likely to gain admission to and succeed in selective STEM programs, i.e., the ones associated with high earnings, and that they are the least affected by strategic concerns in settings that limit the number of programs students can list. This section establishes that the pooled gap is stable to the cutoff, and that where individual settings do move across thresholds they offset one another rather than accumulate. We vary the threshold used to define high-achieving students and the definition of a STEM applicant, and we extend the analysis to students below the top 20%, so that every result in the main text can be read off the full performance distribution rather than a single group.

Appendix B.1. The choice gap is stable across thresholds

Table B.I reports the choice gap at four thresholds, from the top 5% to the top 30%, under both definitions of a STEM applicant. Pooling the ten settings, the first-choice gap moves from −21.5 to −22.4 percentage points and back to −21.9 as the group widens six-fold, a range of 0.9 percentage points; the any-choice gap varies by 0.6. The gap is negative in all forty setting-by-threshold cells under each definition.

The gap widens with the threshold in some settings and narrows in others. Between the top 5% and the top 30% the first-choice gap widens in Taiwan (from −20.7 to −31.9 percentage points) and Australia (from −14.7 to −21.1), and narrows in Brazil, Chile, Finland and Spain; moving from the top 10% to the top 30% it narrows in five of the ten settings. These movements offset rather than accumulate, so the pooled gap is nearly flat. The top-10% column is the joint largest of the four, not one end of a trend. The settings whose gaps move most across thresholds (Taiwan, Greece, Australia) are not the settings with the largest gaps, so the ordering across settings is largely preserved.

Table B.I: The STEM choice gap at alternative performance thresholds

Setting	Top 5%	Top 10%	Top 20%	Top 30%
Panel A. Ranks a STEM program first				
Australia	−14.7	−16.1	−18.9	−21.1
Brazil	−19.7	−20.2	−19.5	−18.9
Chile	−23.1	−23.1	−22.8	−22.3
China	−32.8	−36.5	−38.0	−38.5
Finland	−25.3	−26.2	−25.8	−24.6
Greece	−17.8	−22.5	−24.5	−24.3
Spain	−30.0	−29.7	−29.7	−29.1
Sweden	−23.5	−26.3	−26.0	−25.5
Taiwan	−20.7	−24.7	−28.8	−31.9
Uganda	−12.8	−11.5	−15.6	−17.4
Pooled	−21.5	−22.4	−22.4	−21.9
Range			0.9 percentage points	
Panel B. Ranks a STEM program anywhere on the list				
Australia	−17.0	−17.5	−19.3	−21.1
Brazil	−22.3	−22.7	−22.1	−21.5
Chile	−22.4	−22.7	−22.8	−22.2
China	−28.9	−26.4	−26.1	−25.8
Finland	−20.7	−21.9	−23.8	−24.0
Greece	−23.5	−25.0	−25.7	−25.7
Spain	−29.3	−29.0	−28.8	−28.3
Sweden	−25.1	−27.3	−26.8	−26.8
Taiwan	−6.1	−8.7	−11.0	−13.9
Uganda	−7.4	−9.4	−13.3	−16.4
Pooled	−22.3	−22.8	−22.9	−22.5
Range			0.6 percentage points	

Notes: Female-minus-male difference in the probability of the outcome named in each panel—ranking a STEM program first (Panel A) or anywhere on the preference list (Panel B)—in percentage points, among students at or above each performance threshold. Negative values indicate that women are less likely than men to do so. *Pooled* counts all students in the ten settings together, and *Range* is the spread of the pooled gap across the four thresholds. The main text reports the top 10% column.

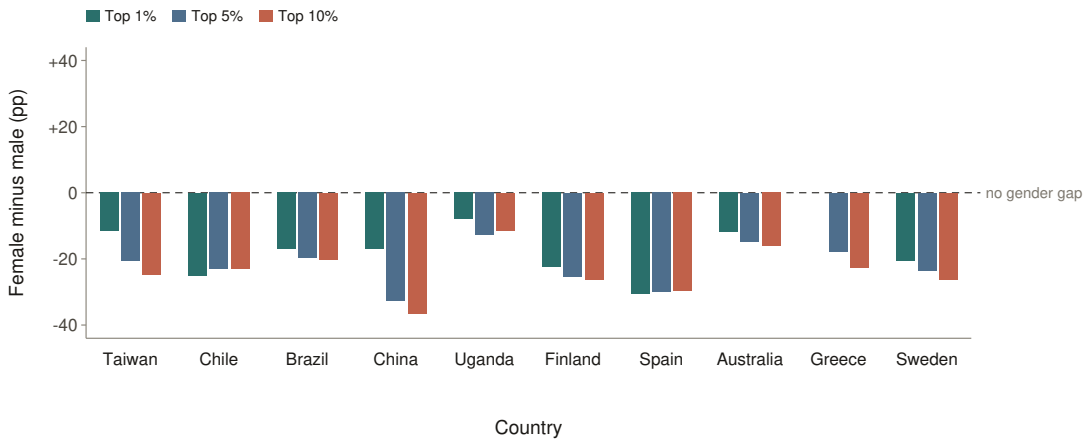
Appendix B.2. The choice gap at the top 1%

Table B.I asks what happens as the high-achieving group is made *wider*. Figure B.I asks the opposite question, and reports the top 1% alongside the top 5% and the top 10%. The group can be formed in nine of the ten settings; Greece is the exception, since its data reach us by decile and top 5%.

The sign is unchanged: the gap is negative in all nine settings at the top 1%, as it is at every other threshold we can construct. The magnitude is not. Averaged over the nine settings the gap falls from −23.8 percentage points at the top 10% to −18.2 at the top 1%, and it is smaller at the top 1% in seven of the nine, with Chile and Spain the exceptions. Two settings account for much of the movement: China falls from −36.5 to −17.0 and Taiwan from −24.7 to −11.5.

The top 1% includes 524 students in China, 547 in Uganda, 695 in Taiwan, and 798 in Australia. As in Table B.I, the choice gap remains negative at every threshold we can construct, including these smaller groups.

Figure B.I: The STEM Choice Gap at the Top 1%, 5%, and 10%

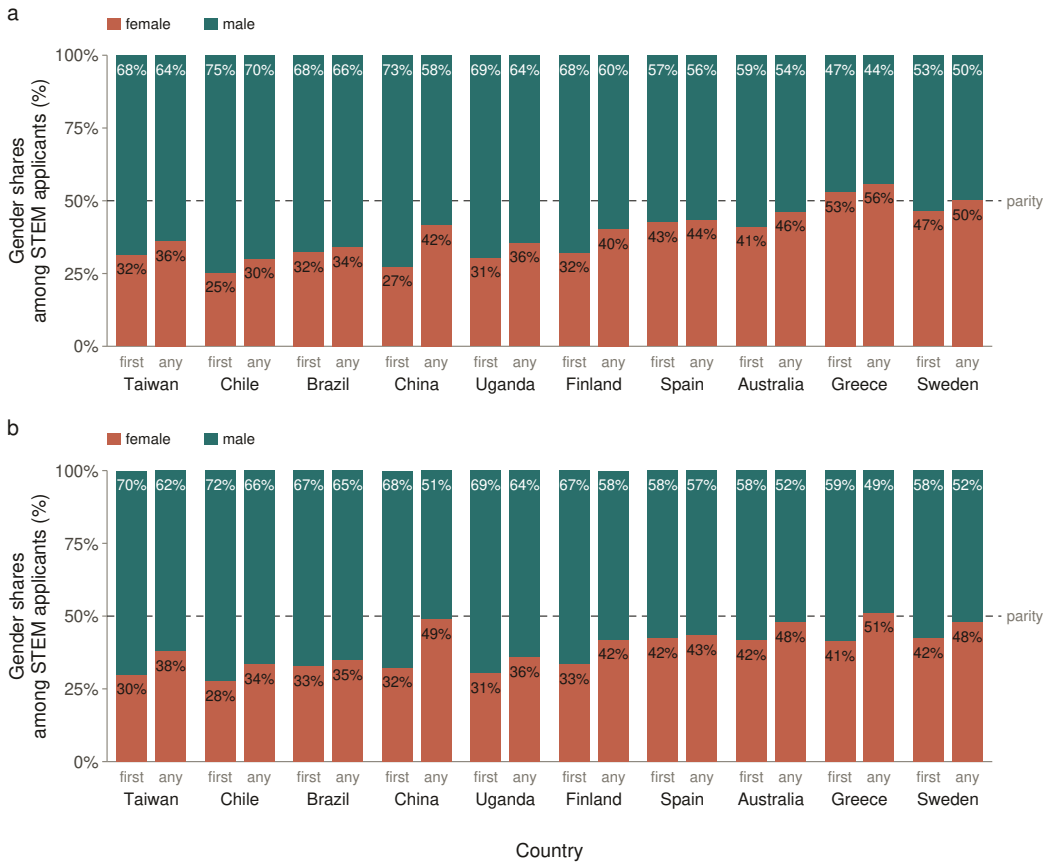


Notes: Female-minus-male difference in the probability of ranking a STEM program first, in percentage points, at three performance thresholds. Greece is omitted from the top 1% because its data are reported by decile and top 5%, so the group cannot be formed. Uganda's top-1% bar is its top score group, 1.5% of applicants, because its seventeen-value admission score admits no exact 1% cutoff (Online Appendix A.1). Denominators are all students in the score group throughout, the convention of Table B.I, with which this figure reconciles exactly; it therefore differs from Figures B.VI and B.VIII, where Chile conditions on applicants. Samples in Online Appendix Table A.II.

Appendix B.3. Gender shares among STEM applicants at alternative performance thresholds

Figure 1 in the main text reports women's share among first-choice STEM applicants, for all applicants and for students in the top 10% of the academic performance distribution. Figure B.II reports the same share at the top 5% and the top 20%, under both the first-choice and the any-choice definitions.

Figure B.II: Gender Shares among STEM Applicants at Alternative Performance Thresholds



Notes: Female (lower) and male (upper) shares among STEM applicants, at the top 5% (a) and the top 20% (b); each bar spans 0–100 and the pair sums to 100 by construction. Within each country the left bar ranks STEM first, the right ranks it anywhere. Samples in Online Appendix Table A.II.

Both thresholds reproduce the main-text picture. Under the first-choice definition, the same six settings—Taiwan, Chile, Brazil, China, Uganda, and Finland—keep women below 35% of high-achieving STEM applicants, and Greece is the only case in which women exceed 50%, at 53.0% among top-5% students, falling to 41.4% at the top 20%. Under the any-choice definition, female shares rise, but the ordering across settings is largely preserved. The largest increases again occur in China, where the share rises by 14.3 percentage points at the top 5% and by 16.6 at the top 20%, remaining below 50% in both cases. Greece is the only setting where women are a majority under this measure at both thresholds, at 55.9% and 51.2%; Sweden reaches 50.3% among top-5% students but falls back to 47.8% at the top 20%.

Appendix B.4. The pipeline gap at alternative performance thresholds

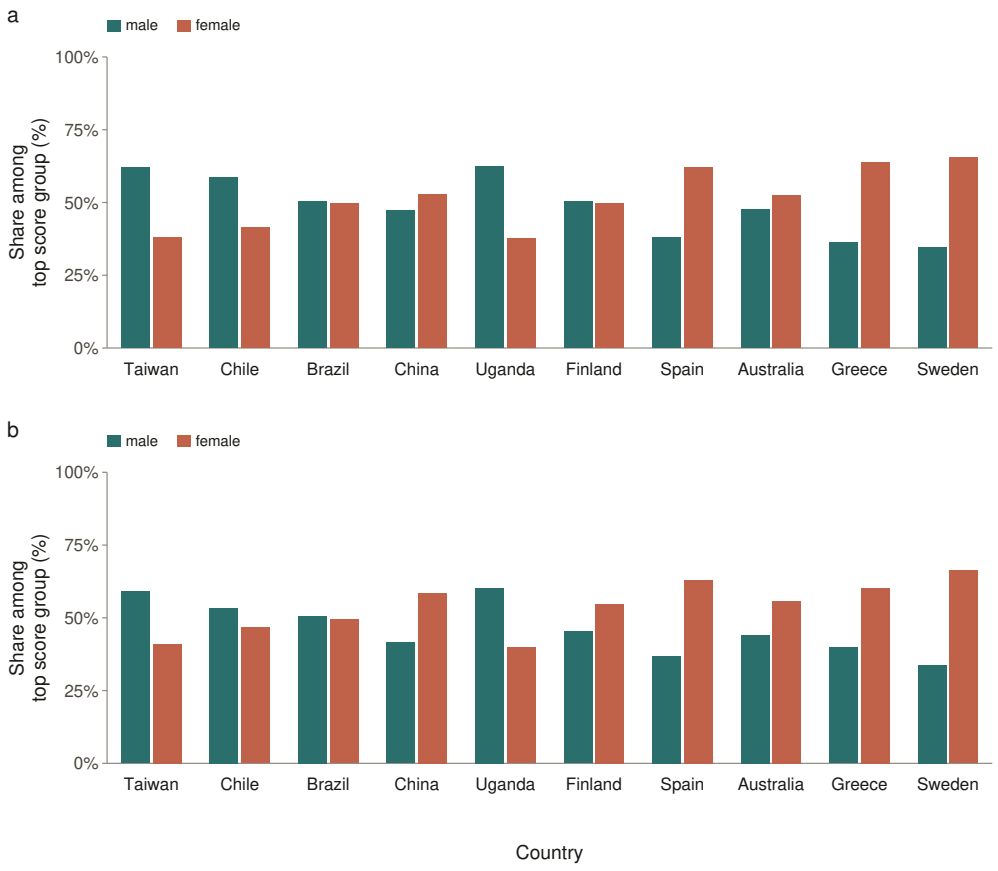
Figure 2 in the main text reports the gender composition of the top 10% of the academic performance distribution and the associated pipeline gap. Figures B.III and B.IV repeat the exercise at the top 5% and the top 20%, the first reporting the shares by sex and the second the gap.

The pattern at both thresholds matches the top decile. Women remain underrepresented among high-achieving students in Taiwan, Chile, and Uganda, and overrepresented in Australia, China,

Greece, Spain, and Sweden, at both thresholds. Brazil stays balanced, with the two shares within about one percentage point of each other everywhere. Finland is the only setting whose gap changes sign between these two thresholds, moving from −0.6 percentage points among top-5% students to +9.3 at the top 20%; Australia widens over the same range, from +4.9 to +11.8. Uganda records the largest negative gap at both thresholds (−24.7 and −20.4 percentage points) and Sweden the largest positive one (+30.7 and +32.6).

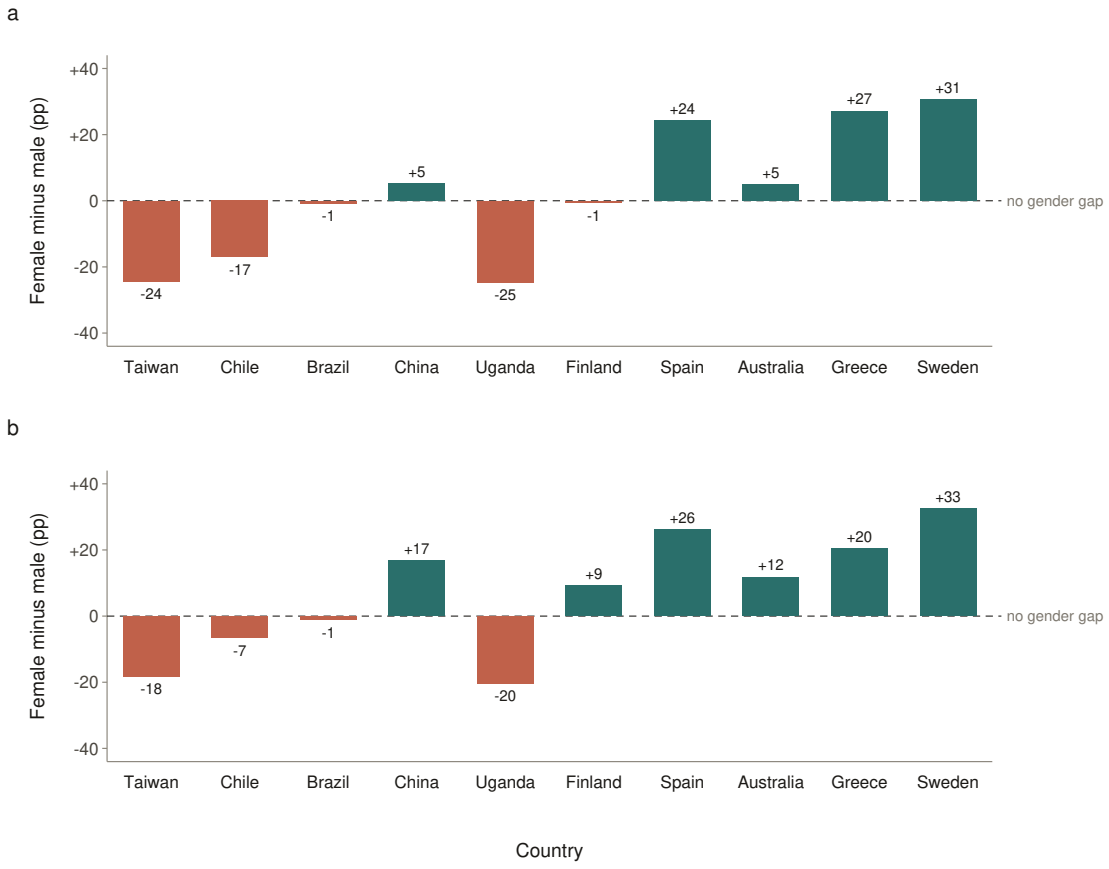
The dispersion persists on the widest group we can form. At the top 30%, the threshold at which Table B.I reports the choice gap, the pipeline gap runs from −20.0 percentage points in Uganda to +31.1 in Sweden, a spread of 51 points on a group three times the size of the top decile. Taiwan, Chile and Uganda remain below parity and Australia, China, Greece, Spain and Sweden above it, as in the main text. The two settings the main text describes as balanced or sign-changing behave accordingly: Finland reaches +13.5, and Brazil, within about one percentage point of parity at every threshold, crosses to +0.3 from −1.1 at the top 20%. The variation in the pipeline gap is a feature of the settings rather than of the cutoff.

Figure B.III: Gender Shares among Top Students at Alternative Performance Thresholds



Notes: Shares of each performance group that are male and female, at the top 5% (a) and the top 20% (b). The female-minus-male pipeline gaps are reported in Figure B.IV. Samples in Online Appendix Table A.II.

Figure B.IV: The Pipeline Gap at Alternative Performance Thresholds



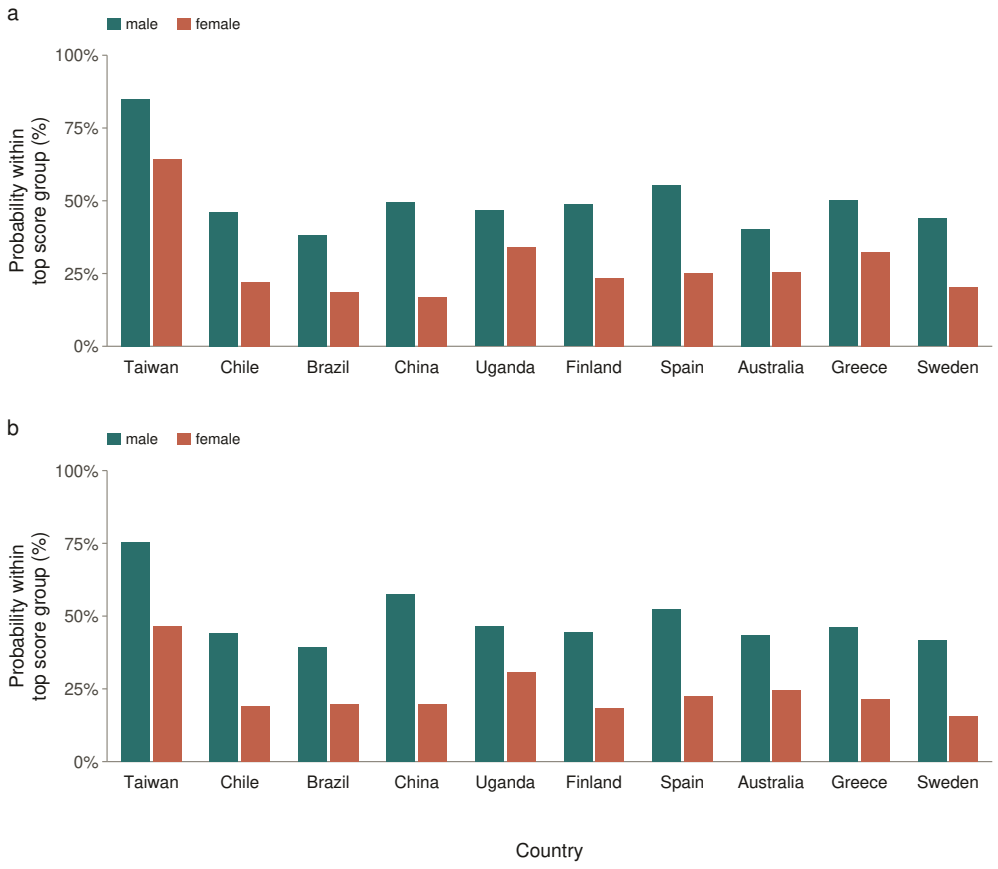
Notes: Female-minus-male pipeline gap, at the top 5% (a) and the top 20% (b). The underlying shares by sex are reported in Figure B.III. Samples in Online Appendix Table A.II.

Appendix B.5. The first-choice STEM gap at alternative performance thresholds

Figure 2 in the main text reports the share of top-10% women who rank a STEM program first and the associated gender gap. Figures B.V and B.VI repeat the exercise at the top 5% and the top 20%, the first reporting the probabilities by sex and the second the gap.

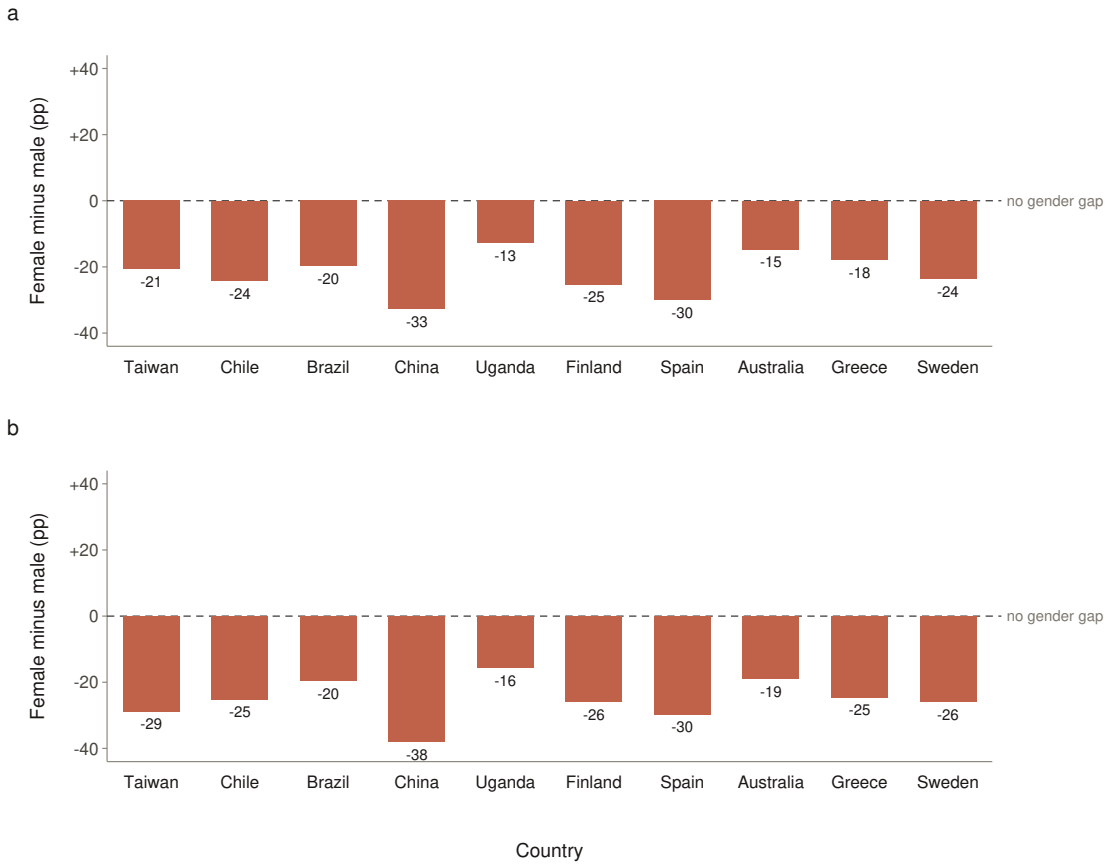
The gap is negative in all ten settings at both thresholds, as it is at the top 10%, and it exceeds 20 percentage points in six of the ten among top-5% students and in seven at the top 20%, against eight at the top 10%. Uganda records the smallest gap at both thresholds (−12.8 and −15.6 percentage points) and China the largest (−32.8 and −38.0). The ordering across settings is broadly stable; the clearest exception is Taiwan, where the gap widens from −20.7 percentage points among top-5% students to −28.8 at the top 20%, moving it from the sixth to the third largest.

Figure B.V: STEM First-Choice Probabilities by Sex at Alternative Performance Thresholds



Notes: Probability of ranking a STEM program first, by sex, at the top 5% (a) and the top 20% (b). Chile conditions on applicants. The female-minus-male gaps are reported in Figure B.VI. Samples in Online Appendix Table A.II.

Figure B.VI: The STEM First-Choice Gender Gap at Alternative Performance Thresholds



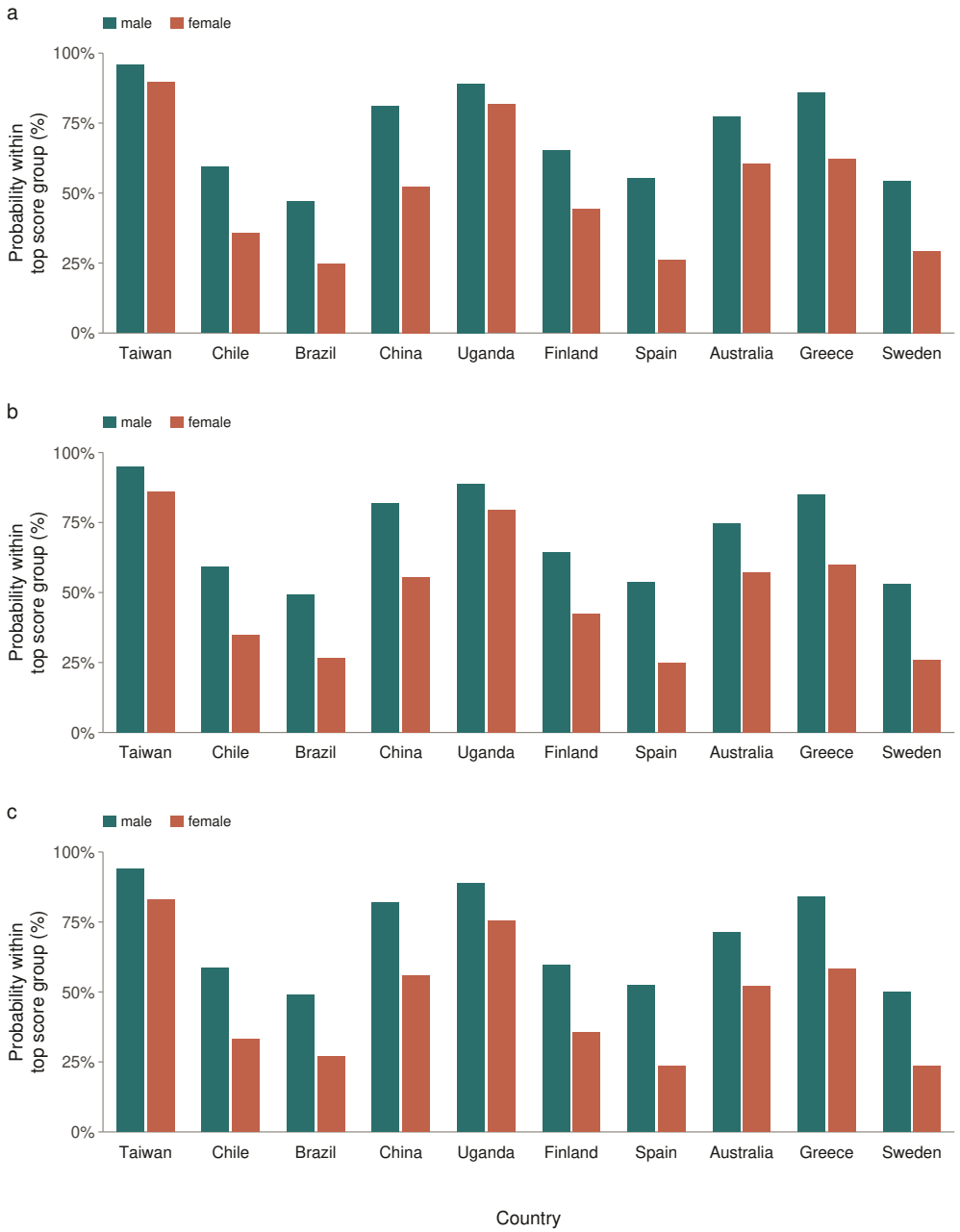
Notes: Female-minus-male gap in the probability of ranking a STEM program first, at the top 5% (a) and the top 20% (b). Chile conditions on applicants. The underlying probabilities by sex are reported in Figure B.V. Samples in Online Appendix Table A.II.

Appendix B.6. The choice gap under an any-choice definition of a STEM applicant

The main text measures the choice gap using students who rank a STEM program first, the stricter of the two definitions. Figures B.VII and B.VIII repeat the exercise under an any-choice definition, which counts a student as a STEM applicant whenever a STEM program appears anywhere on the application list, at all three performance thresholds: the first reports the probabilities by sex, the second the female-minus-male gap.

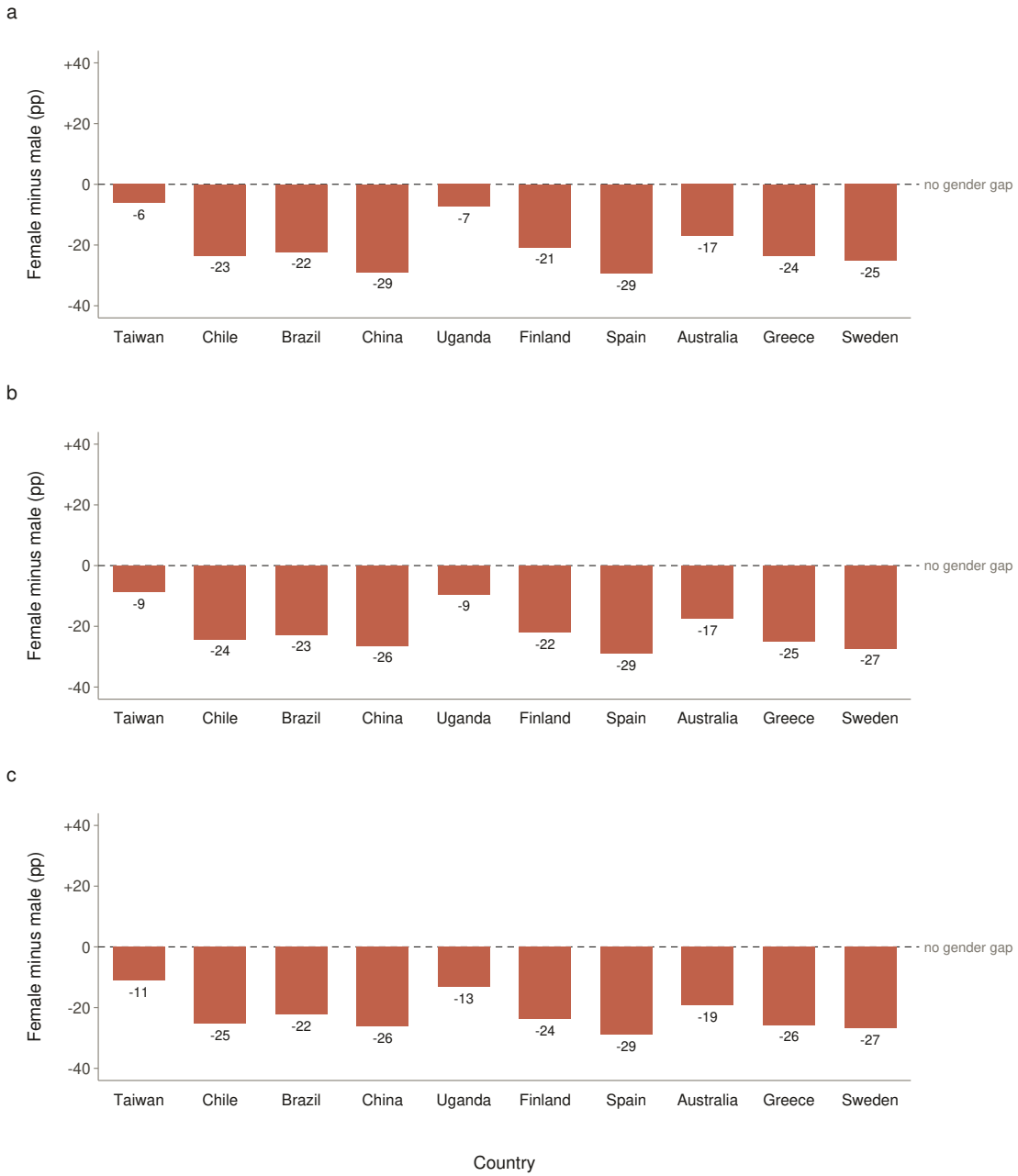
The gap is negative in all ten settings at every threshold, as under the first-choice measure. Averaged across settings it is smaller than under the first-choice measure, though pooling students it is marginally larger (-22.8 against -22.4 at the top 10%), and it is larger under the any-choice definition in Australia, Brazil, Greece and Sweden. Under either aggregation it remains substantial: it exceeds 20 percentage points in seven of the ten settings at each of the three thresholds. The three smallest gaps are found in Taiwan, Uganda, and Australia at all three thresholds, with Taiwan the smallest throughout (-6.1, -8.7, and -11.0 percentage points). No setting changes its position markedly across thresholds.

Figure B.VII: Any-STEM Application Probabilities by Sex at Alternative Performance Thresholds



Notes: Probability of listing a STEM program anywhere, by sex, at the top 5% (a), top 10% (b) and top 20% (c). Chile conditions on applicants. The female-minus-male gaps are reported in Figure B.VIII. Samples in Online Appendix Table A.II.

Figure B.VIII: The Any-STEM Gender Gap at Alternative Performance Thresholds



Notes: Female-minus-male gap in the probability of listing a STEM program anywhere, at the top 5% (a), top 10% (b) and top 20% (c). Chile conditions on applicants. The underlying probabilities by sex are reported in Figure B.VII. Samples in Online Appendix Table A.II.

Appendix B.7. Across the full score distribution

To extend the analysis below the top quintile, we pool deciles 1–5 into a single "bottom 50%" group and report deciles 6–10 separately. This preserves a decile-by-decile view in the upper tail,

where admission to selective STEM programs is most plausible, while reducing noise in the lower half of the distribution.

Female representation tends to decline with academic performance in Brazil, Chile and Finland, and to increase in Greece and Sweden. The female-minus-male gap in STEM application probabilities is negative in all 120 country-by-bin cells—ten settings, six score bins, and two definitions of a STEM application. Its magnitude varies with performance in some settings, but its sign does not.[23]

Appendix B.8. Counterfactual female shares among STEM assignees

The main text reports that the assignment component V is small and carries no systematic sign, and Figure 4 there compares the gender composition of the four groups. Table B.II gives the setting-by-setting counterfactuals behind that statement.

The assignment margin moves the female share by less than two percentage points in seven of the ten settings and is positive in five. The largest exception is Uganda, at -5.7 percentage points. Uganda also has the lowest ratios of STEM assignments to STEM first preferences, 0.49 for men and 0.37 for women. These count ratios are not acceptance probabilities: assignments can come from lower-ranked STEM choices. Uganda's admission rules differ explicitly by sex (Online Appendix A.1); elsewhere we document no sex-conditional rule (Table A.III). Net assignment differences can reflect program rankings, performance within the top group, and entry from lower list positions. The aggregate component does not isolate these channels or test treatment at a common cutoff.

Table B.II sets the two counterfactuals implied by (E.1) side by side. Giving high-achieving women men's rate of STEM assignment raises the female share among STEM assignees in all ten settings, in most cases substantially: from 42.2% to 65.9% in Sweden and from 31.5% to 49.4% in Brazil. This equalizes the combined choice and assignment margin ($C + V$), not the choice component alone. Balancing the top decile while leaving each gender's assignment rate as observed raises the share in only four settings. In the other six—Australia, China, Finland, Greece, Spain and Sweden—it *lowers* it, by 14.7 percentage points in Sweden, 12.3 in Spain and 10.7 in Greece. The reason is arithmetic rather than behavioral: women already outnumber men among high achievers in those settings, and that surplus is currently offsetting part of the choice gap, so removing it removes the offset as well. Neither column predicts how a margin would respond to an intervention.

[23]The three figures behind these statements—female shares by score bin, and the first-choice and any-choice gaps by bin, for every setting—are reported in the project's online materials at https://gendergapinsights.com, together with the underlying counts.

Table B.II: Counterfactual female shares among high-achieving students assigned to STEM

		Counterfactual		Change from
Setting	Observed	Balanced pipeline	Common STEM assignment rates	balancing
Uganda	26.3	36.2	38.6	+9.9
Chile	27.5	32.5	44.0	+5.1
Taiwan	29.3	38.8	39.5	+9.5
Brazil	31.5	32.0	49.4	+0.5
China	33.3	28.8	55.3	−4.6
Finland	35.1	33.6	51.6	−1.5
Sweden	42.2	27.5	65.9	−14.7
Australia	42.8	39.2	53.7	−3.6
Spain	44.8	32.5	62.8	−12.3
Greece	45.1	34.4	61.1	−10.7

Notes: All entries are the percentage of high-achieving students assigned to a STEM program who are women, where high-achieving means the top decile of each setting's academic performance distribution. *Observed* is the realized share. *Balanced pipeline* holds each gender's rate of reaching a STEM assignment at its observed value and sets the top decile to equal numbers of women and men; it equals $1-\Lambda(C+V)$ in the notation of (E.1) (Online Appendix E). *Common STEM assignment rates* holds the observed top-decile composition and gives women men's assignment rate M_m/T_m, setting $C+V=0$; the female share is then $1-\Lambda(P)$. The final column is the balanced-pipeline share minus the observed share; a negative entry means that balancing the pipeline, on its own, would lower the female share among STEM assignees, because women already outnumber men in that setting's top decile. Both columns are accounting exercises that reallocate observed rates and are not predictions of how either margin would respond to an intervention.

Appendix C. The choice gap in other selective degrees

Table C.I compares STEM with health and business/law on a common log-ratio scale. The summaries below report female-minus-male first-choice gaps at the top 5%, 10%, and 20%. The corresponding probabilities, any-choice gaps, figures, and counts are available at https://gendergapinsights.com.

Appendix C.1. Business and law

At the top 10%, the gap is negative in four of nine settings and positive in five, averaging +2.4 percentage points and ranging from −13.5 in Australia to +14.5 in Taiwan. Taiwan exceeds +13 points at every threshold, and China at the top 5% and 10% (+14.2 and +13.5), easing to +11.9 at the top 20%; Australia ranges from −10.7 to −13.5. Chile, Brazil, and Finland are mildly negative, Spain and Sweden mildly positive. No setting changes sign across thresholds: the unweighted means are +2.9, +2.4, and +2.5 points. Uganda moves most, from +4.1 to +11.1. Greece reports business/law only at the top 5% (+9.0).

Appendix C.2. Health

The gap is positive in eight settings at the top 5% and 10%, and nine at the top 20%, averaging +8.2 points at each threshold. Chile, Brazil, Finland, and Spain range from +13 to +20 points; Sweden from +8.9 to +10.6. Taiwan is negative throughout, by at most 2.5 points. Greece moves from −6.6 to −1.9 and then marginally positive. China and Uganda sit close to zero at the top 10%; Uganda moves from +9.8 to +1.4 as the group widens.

These comparisons hold the applicant pool, performance threshold, admission mechanism, and list-length constraint fixed within each setting. The consistently negative STEM gap is field-specific.

Table C.I: Decomposition of the gender gap among high-achieving applicants and assignees, by field

	Pipeline	Choice component C			Assignment	Balanced-pipeline
Setting	P	STEM	Health	Business/Law	V_{STEM}	male share $\Lambda(C_{\text{STEM}})$
Australia	−0.15	+0.48	−0.25	+0.64	−0.05	0.62
Brazil	+0.02	+0.70	−0.60	+0.06	+0.05	0.67
Chile	+0.24	+0.77	−0.67	+0.12	−0.04	0.68
China	−0.21	+1.07	−0.10	−0.74	−0.17	0.75
Finland	−0.07	+0.78	−0.52	+0.40	−0.10	0.69
Greece	−0.45	+0.64	+0.08	–	+0.01	0.65
Spain	−0.52	+0.81	−0.46	−0.30	−0.08	0.69
Sweden	−0.66	+0.91	−0.40	−0.11	+0.06	0.71
Taiwan	+0.43	+0.37	+0.04	−0.83	+0.08	0.59
Uganda	+0.46	+0.29	−0.06	−0.88	+0.28	0.57
Positive in	4/10	10/10	2/10	4/9	5/10	

Notes: High-achieving students are those in the top decile of each setting's academic performance distribution. P is the pipeline component and C the choice component of (1); V is the assignment component of (E.1), which extends the identity to realized assignments. Positive values favor men, and P is common to all fields within a setting by construction. $\Lambda(C)$ is the male share among high-achieving STEM applicants that would obtain if the pipeline were balanced. Online Appendix E derives both identities and reports V for the other fields. Greece does not report business/law counts. Rates here use each setting's full top-decile totals as denominators, so that the components satisfy the identity exactly; Chile's panel (b) of Figure 2 instead conditions on students who submit an application, which is why Chile's C and its plotted gap are not directly comparable.

Appendix D. Pipeline and STEM choice gaps in cohorts closest to 2018

Appendix D.1. Sample and time coverage

This appendix evaluates whether the main findings are sensitive to cohort composition by re-estimating top-10% pipeline and choice gaps using cohorts as close as possible to 2018 in each setting. The objective is to make the time period as comparable as possible across countries, although this is not feasible everywhere given differences in the underlying source files.

The country-year selections used are: Australia (2010), Brazil (2016), Chile (2018), China (2018), Finland (2018), Greece (2003–2012 pooled), Spain (2020), Sweden (2017), Taiwan (2010), and Uganda (2017/2018).

Relative to the main-text samples, we move closer to 2018 in Australia (from 2009–2010 pooled to 2010), Chile (from 2012–2018 pooled to 2018), Finland (from 2016–2020 pooled to 2018), and Spain (from 2020–2024 pooled to 2020). In the remaining settings we use the same period as in the main text because no closer harmonized cohort is available in the current country files. China (2018), Sweden (2017), Brazil (2016), and Uganda (the 2017/2018 academic year, entering in fall 2017) are close to the target year; Taiwan (2010) and Greece (2003–2012) are further from it, as is Australia (2010) even on its narrowed window, so the exercise cannot address period differences for those three settings.

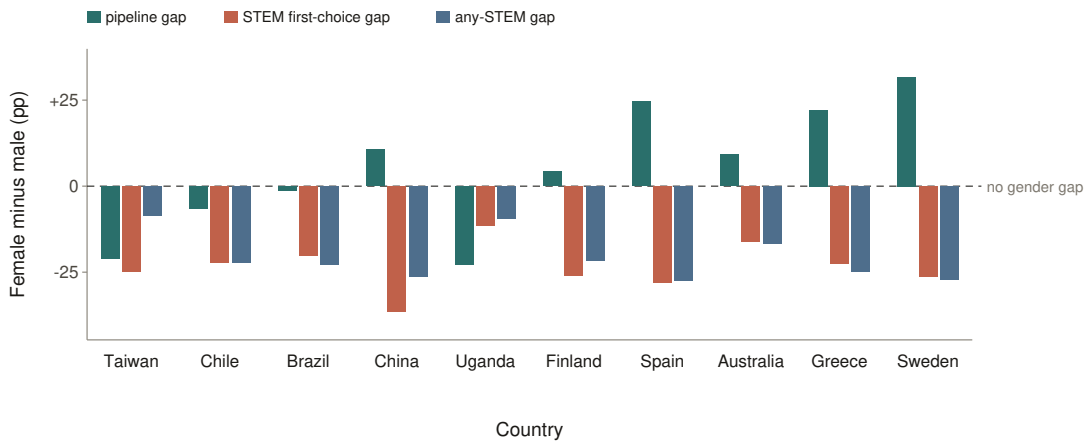
Appendix D.2. Pipeline gap in recent cohorts

In Figure D.I, the first series shows the top-10% pipeline gap for cohorts closest to 2018. As in the main text, the pipeline gap continues to vary substantially across countries and changes sign across settings.

Appendix D.3. Choice gaps in recent cohorts

The second and third series in Figure D.I report the first-choice and any-choice STEM gaps in recent cohorts. The first-choice STEM gap remains negative and sizable in all countries, and the any-choice STEM gap is also negative in all countries, with magnitudes generally close to the first-choice benchmark. Restricting the sample to cohorts closest to 2018 leaves both patterns unchanged.

Figure D.I: Top-10% Pipeline and Choice Gaps in Cohorts Closest to 2018



Notes: Female-minus-male gaps for the top 10%, using the cohort closest to 2018 in each setting. Online Appendix D.1 lists the country-years used.

Appendix E. The pipeline–choice decomposition

The main text states the identity in equation (1): at a fixed performance threshold the pipeline component P and the choice component C sum exactly to the gender gap among high-achieving applicants to a field. This appendix extends it to realized assignments and reports the additional component.

Extending the identity to assignment. Equation (1) stops at the application stage. Because each of our settings places students through a centralized assignment mechanism, we also observe where the same students end up, and the identity extends by one further factor. Let M_g denote the number of students of gender g in the top group assigned to the field, and let $v_g = M_g/A_g$. Then

$$\underbrace{\ln\left(\frac{M_m}{M_f}\right)}_{\text{gap among high-achieving assignees}} = P + C + \underbrace{\ln\left(\frac{v_m}{v_f}\right)}_{\text{assignment component } V}, \tag{E.1}$$

so that the male share among high-achieving students assigned to the field is $\Lambda(P+C+V)$. The third factor is a ratio of counts rather than a conditional probability: a student may be assigned to a field she did not rank first, and v_g exceeds one for at least one gender in four of our settings. V therefore combines two channels, differential admission among those who ranked the field first and differential entry from lower positions in the submitted list.

In our data V is small and carries no systematic sign. Across the ten settings it ranges from -0.17 to $+0.28$ log points, lies below 0.10 in absolute value in eight, and is smaller in magnitude than C in nine settings and comparable in the tenth (Uganda, 0.28 against 0.29); its mean is $+0.005$. The assignment component is similarly small in the other fields, ranging over $[-0.21, +0.06]$ for health and $[-0.46, +0.12]$ for business/law.

Its magnitude is stable across thresholds: recomputing V on the top 20% moves no setting by more than 0.12 log points. The gender composition of assigned cohorts is therefore close to that implied by submitted application lists. This aggregate result does not establish that the mechanism treats otherwise identical men and women alike.

We report P, C and V in log points rather than as shares of the total. The components frequently carry opposite signs in our data, and ratios of the form $P/(P+C)$ are not well behaved in that case. Equations (1) and (E.1) are accounting identities: they partition the observed gap into measured components but do not identify their causes.

Appendix F. From applications to assignments

Figure 4 compares the gender composition of four groups of high achievers, from the top decile to those assigned to STEM. Online Appendix E extends (1) to realized assignments and defines the assignment component V in (E.1).

V is small and carries no systematic sign: $|V|$ is at most 0.10 in eight of the ten settings, smaller in magnitude than C in nine of the ten settings and comparable in the tenth (Uganda, 0.28 against 0.29), and averages +0.005. Because V is small, the counterfactual in the final column of Table C.I carries over from applicants to assignees: with a gender-balanced top group, the male share among high-achieving students assigned to STEM would lie between 0.61 and 0.73. Uganda operates admission rules that differ by sex and favor women (Online Appendix A.1); its assignment component nonetheless runs against women, a pattern these data cannot explain and which we leave as an open question. Online Appendix B.8 reports the counterfactuals implied by (E.1) setting by setting.

Appendix G. The pipeline gap, the choice gap, and gender wage parity

This section reports the cross-country evidence behind the discussion in Section 4.6 of the main text. Figure 5 relates the *pipeline* and *choice gaps* among students in the top 10% of the academic performance distribution to the wage parity index, defined as one minus the total gender wage gap, so values closer to one imply greater parity: panel (a) reports the pipeline gap and panel (b) the choice gap. Wage gaps come from the International Labour Organization (2026), except for Australia and Taiwan, taken from official sources (Workplace Gender Equality Agency, 2026, Directorate-General of Budget, Accounting and Statistics (DGBAS), Taiwan, 2016), and China, from Zhang and Dong (2025).

Appendix G.1. Sensitivity of the two slopes

In log-ratio units, the unweighted slopes per unit of wage parity are -2.25 for P ($t = -3.7$) and $+1.02$ for C ($t = +2.0$). These components use full top-decile totals, including Chile, as in Table C.I. Figure 5 instead plots female-minus-male percentage-point gaps and, like Figure 2, conditions Chile's application rates on submitted lists. Pipeline precision is similar on either scale ($|t| = 3.70$ versus 3.71). Leave-one-setting-out slopes on C range from +0.58 to +1.22, whereas the percentage-point slope changes sign. Excluding Uganda lowers the statistic for C to $t = 0.7$. The persistent choice gap is robust across settings; the direction and precision of its association with wage parity are sensitive to scale and sample composition.